Investigation of Compressible Flow Formulation in the Low- Mach, Low-Reynolds Regime

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Objective:

The objective of this study is to investigate whether the compressible flow equations implemented in SU2 can accurately predict flow behaviour at low Reynolds numbers (Re = 10,000 and below). In such regimes, the corresponding Mach numbers become very small, approaching the incompressible limit. Many numerical solvers designed for compressible flows tend to lose accuracy or stability under these conditions. Therefore, this study aims to assess the applicability and limitations of the compressible formulation in SU2 for low-Reynolds-number flows.

Nomenclature:

α:	Speed pf sound	M:	Mach number
γ:	Ratio of specific heats	V:	Velocity
ρ:	Density	R:	Gas constant
τ_w:	the wall shear stress	Re:	Reynolds number
μ:	Dynamic viscosity	S:	Sutherland's constant
μ_{ref}:	Sutherland's Reference dynamic viscosity	T:	Freestream temperature
ν:	Kinematic viscosity	T_{ref}:	Sutherland's Reference temperature
C_L:	Left coefficient	u_τ:	The friction velocity
C_D:	Drag coefficient	y:	Distance of first cell centre
L:	Characteristic length	$y+$:	Wall units

Introduction:

At low Mach numbers, the large disparity between acoustic and convective wave speeds leads to stiffness and poor conditioning of the compressible Navier–Stokes equations, resulting in degraded convergence and potential inaccuracies (Turkel, 1999). This behaviour poses a significant challenge for numerical solvers that are primarily designed for compressible flow regimes.

However, an important question arises regarding the limits of applicability of these formulations: at what point do compressible solvers cease to provide reliable results as the Mach number decreases? For low-Reynolds-number flows—where velocities are typically small and the flow approaches the incompressible limit—it becomes essential to evaluate whether compressible formulations remain accurate and efficient, or whether alternative approaches are required.

To investigate these limitations, a specific test case is considered. The flow around a circular cylinder at low Reynolds numbers is analysed, and the corresponding flow properties are determined. The following sections outline the calculations used to define the flow conditions.

As an upper bound for the present study, a Reynolds number of $Re = 10{,}000$is selected. At this limit, it is first necessary to evaluate the corresponding Mach number to assess the relevance of compressibility effects.

Given:

$L = 1m$

$V = 1\ m/s$

$\rho = 1\ kg/m^3$

Using the Reynolds number definition:

$$Re = \frac{(\rho\ V\ L)}{\mu} \qquad (1)$$

Solving for dynamic viscosity one gets:

$$\mu = \frac{(\rho\ V\ L)}{Re}$$

$$\mu = \frac{(1 \times 1 \times 1)}{10000} = 1 \times 10^{-4}\ Pa \cdot s$$

Using Sutherland's law:

$$\mu = \mu_{ref} \left(\frac{T}{T_{ref}}\right)^{\frac{3}{2}} \cdot \frac{(T_{ref} + S)}{(T + S)} \qquad (2)$$

Where for air at experiment conditions, reference values are:
$\mu_{ref} = 1.716 \times 10^{-5} Pa \cdot s$
$T_{ref} = 273.15\ K$

$S = 110.4\ K$

Substituting values in (2) above:

$$1 \times 10^{-4} = 1.716 \times 10^{-5} \times \left(\frac{T}{273.15}\right)^{\frac{3}{2}} \times \frac{273.15 + 110.4}{(T + 110.4)}$$

Solving for T:

$$T \approx 1130\ K$$

Now substituting the temperature value in the Mach number definition (3) and (4) below:

$$M = \frac{V}{\alpha} \qquad (3)$$

$$\alpha = \sqrt{\gamma R T} \qquad (4)$$

Where for air:

$\gamma = 1.4$

$R = 287\ \mathrm{J/kg.K}$

$$\alpha = \sqrt{1.4 \times 287 \times 1130} = 674\ \mathrm{m/s}$$

$$M = \frac{1}{674} = 0.00148$$

A parametric study is conducted by progressively reducing the Mach number from $M = 0.1$ to 0.001. This approach allows a more detailed assessment of compressibility effects and their influence on the flow behavior.

Mesh:

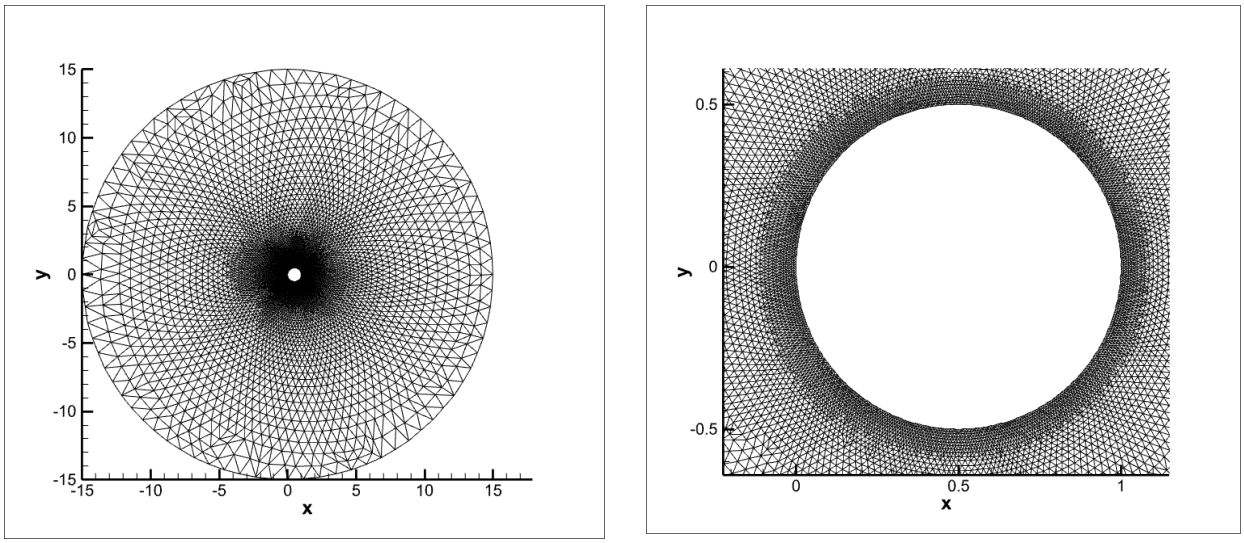

Figure 1: The figure on the left shows the full mesh, while the figure on the right provides a zoomed-in view of the boundary region near the cylinder.

Note: All the cases have a Reyolds number and a characteristic length of:

$Re = 40$

$L = 1\,m$

The flow around a circular cylinder at this Reynolds number is well documented and is known to be steady, with no vortex shedding present. As a result, the coefficients $C_L$and $C_D$reach steady values without oscillations, eliminating the need for RMS calculations or other time-averaging techniques. Thus it was selected.

Setup:

Case 1:

Given:

$M = 0.1$

$T = 288.15\ \mathrm{K}$

Note: In this case, the temperature is fixed to ensure closure of the system and allow the determination of the remaining flow variables.

In SU2, it is not necessary to explicitly specify the velocity V, dynamic viscosity μ, or density ρ, as these quantities can be computed directly from the Mach number relation, Sutherland's law, and the Reynolds number. The following calculations are therefore carried out to determine the flow variables for Cases 2 and 3.

Substituting equation (4) into (3) and solving for V:

$$V = M \cdot \sqrt{\gamma R T} \qquad (5)$$

$$V = 0.1 \cdot \sqrt{1.4 \times 287 \times 288.15} = 34\, m/s$$

Since SU2 uses Sutherland's law by default to model viscosity, the dynamic viscosity is calculated using Equation (2).

$$\mu = 1.716 \times 10^{-5} \left(\frac{288.15}{273.15}\right)^{\frac{3}{2}} \cdot \frac{(273.15 + 110.4)}{(288.15 + 110.4)} = 1.79 \times 10^{-5}\ \mathrm{Pa \cdot s}$$

Now using Reynolds number definition (equation 1), we can solve for density to get:

$$\rho = \frac{(Re \cdot \mu)}{(V\ L)}$$

$$\rho = \frac{(40 \times 1.79 \times 10^{-5})}{(34 \times 1)} = 2.105 \times 10^{-5}\ \mathrm{kg/m^3}$$

Note: From physics point of view, this corresponds to a Not standard air anymore, but instead a very rarefied gas (almost vacuum).

Case 2:

Assumptions:

M = 0.01

In this case, the density and viscosity values obtained from Case 1 are used to determine the remaining flow variables.

ρ = $2.105 \times 10^{-5}\mathrm{kg/m^3}$

μ = $1.79 \times 10^{-5}\mathrm{Pa \cdot s}$

Note: In this approach, the density $\rho$and dynamic viscosity μ are maintained constant to preserve the mesh configuration. This guarantees that the y^+values remain unchanged, as they are directly dependent on these parameters. Consequently, the Mach number is controlled by adjusting the temperature. As shown in equations (6) and (7) below, since y^+depends on $\rho$and μ, variations in temperature do not influence the near-wall resolution or the associated computations.

The distance from the wall measured in viscous lengths - or wall units -is denoted by:

$$y+ = \frac{u_\tau y}{\nu} \qquad (6)$$

Where the friction velocity is given by:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \qquad (7)$$

The goal now is to calculate T, using Reynolds number definition equation (1), we can rewrite the equation to solve for V:

$$V = \frac{(Re\,.\,\mu)}{(\rho.\,L)}$$

Let the subscript "1" denote the parameters corresponding to Case 1, and the subscript "2" denote those corresponding to Case 2. since:

$$Re_1 = Re_2$$

$$L_1 = L_2$$

$$\rho_1 = \rho_2$$

$$\mu_1 = \mu_2$$

It can be observed that the velocity remains unchanged, i.e., $V_1 = V_2$. From Equation (5), we obtain:

$$M_1 \cdot \sqrt{\gamma_1\, R_1\, T_1} = M_2 \cdot \sqrt{\gamma_2\, R_2\, T_2}$$

Since the gas is air in both cases:

$$\gamma_1\, R_1 = \gamma_2\, R_2$$

Solving for T_2 :

$$T_2 = \left(\frac{M_1}{M_2}\right)^2 T_1$$

$$T_2 = \left(\frac{0.1}{0.01}\right)^2 288.15 = 28815\, K$$

Following the same procedure, the calculations are performed for $M = 0.001$, yielding $T_3 = 2{,}881{,}50$. The results for all cases are summarized in Table (1) below.

Case	$\boldsymbol{M}$	$\boldsymbol{V}$ $\mathbf{m/s}$	$\boldsymbol{\rho}$ $\mathbf{kg/m^3}$	$\boldsymbol{\mu}$ **Pa.s**	$\boldsymbol{T}$ **K**
1	0.1	34	2.105×10^{-5}	1.79×10^{-5}	288.15
2	0.01				288,15
3	0.001				288,150,0

Table 1: Summary of the computed free-stream parameters

Results:

Case1:

This case serves as a validation (sanity check) for the subsequent simulations. As expected, see Figure (2), at this low Reynolds number no vortex shedding occurs, and both $C_L$and $C_D$converge to steady-state values.

From Table (2), the computed $C_L$remains within the commonly accepted numerical tolerance for CFD simulations and is therefore consistent with the expected zero-lift condition. This small deviation from zero lift can be attributed to numerical and discretization effects rather than physical asymmetry. In particular, the use of an unstructured mesh introduces slight geometric and topological asymmetries. Unlike structured meshes, perfect symmetry about the centreline cannot be guaranteed, which may lead to minor

imbalances in pressure and shear stress distributions. The predicted $C_D$values are also in good agreement with reference data ($C_D = 1.48$).

As shown in Figure (3), the pressure contours exhibit the expected distribution for flow around a circular cylinder. The maximum pressure occurs at the stagnation point on the upstream face of the cylinder, where the flow velocity approaches zero. In contrast, the minimum pressure values are observed along the sides of the cylinder, where the flow accelerates and the velocity reaches its maximum.

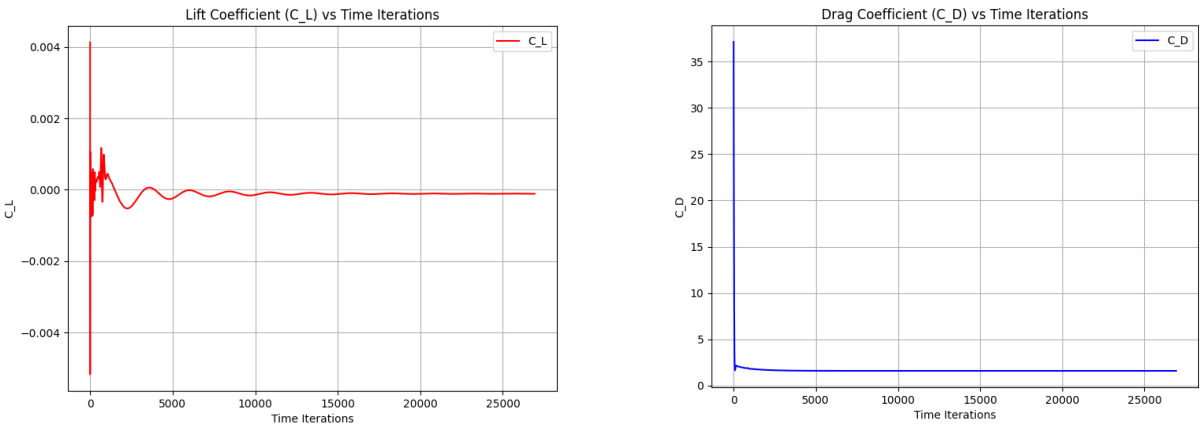

Figure 2: Lift coefficient (left) and drag coefficient (right) at M=0.1

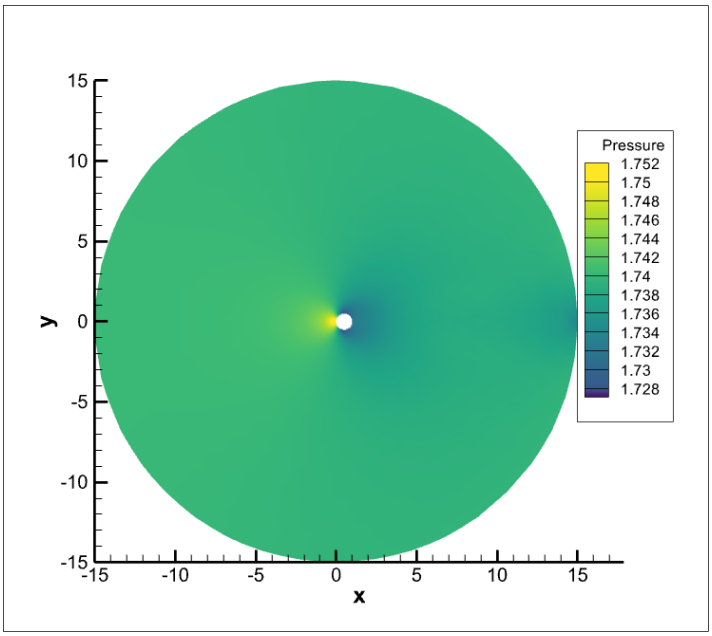

Figure 3: Pressure contour at M = 0.1

Case2:

As shown in Figure (4), the $C_D$coefficient reaches a steady value at an early stage, while $C_L$exhibits small-amplitude oscillations around zero.

For Mach numbers below approximately $M = 0.3$, density variations remain small (typically less than 5%), and the flow can be considered incompressible. In this regime, the governing equations reduce to the incompressible Navier–Stokes equations, and aerodynamic coefficients such as $C_D$and $C_L$depend primarily on the Reynolds number and geometry, rather than on the Mach number.

Therefore, the values of $C_D$and $C_L$are expected to be consistent with those obtained for $M = 0.1$(Case 1). However, a noticeable discrepancy is observed. In particular, the computed $C_D$deviates from the value obtained in Case 1 by approximately 6.25%. While the magnitude of $C_L$increases by approximately one order of magnitude, indicating a significant deviation from the expected near-zero lift condition.

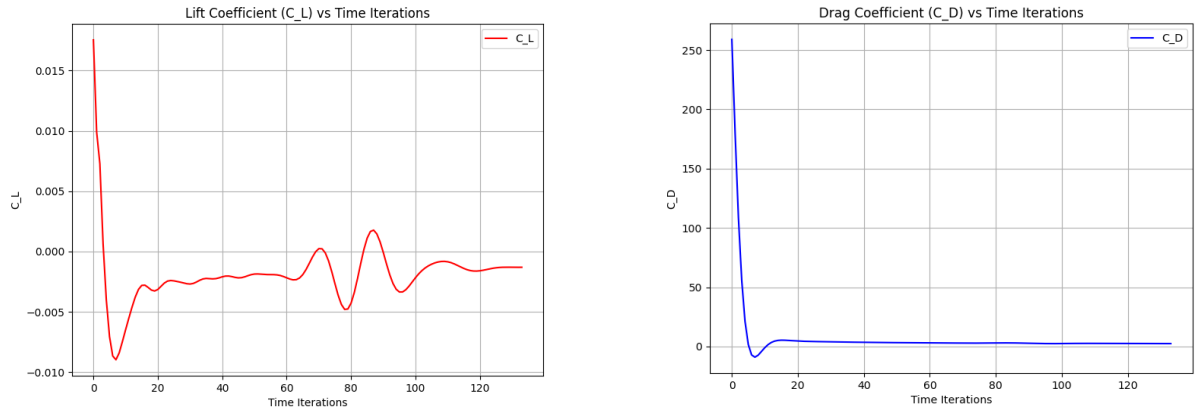

Figure 4: Lift coefficient (left) and drag coefficient (right) at M=0.01

Case 3:

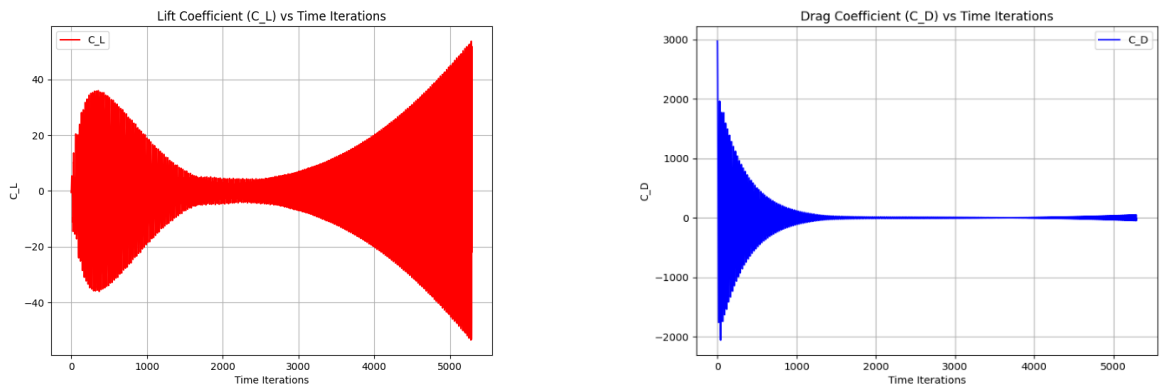

Figure 5: Lift coefficient (left) and drag coefficient (right) at M=0.001

As shown in Figure (5), at this Mach number the solver fails to capture the correct flow physics, and the values of $C_L$and $C_D$diverge.

Conclusion:

The observed differences in $C_L$and C_D in case 1 and case 2are not attributable to changes in the Reynolds number or viscous scaling, but rather to the use of a compressible solver at very low Mach numbers. At very low Mach numbers, compressible solvers are known to suffer from numerical stiffness and inaccurate pressure–velocity coupling, which can lead to non-physical pressure fields and, consequently, deviations in the computed aerodynamic coefficients.

Table (2) summarizes the C_L and C_D values of the last time step of the 3 cases.

Case	C_L	C_D
1	-0.00011	1.602993
2	-0.00217	1.7
3	Solution diverges	

Table 2: A summary of C_L and C_d values of all cases

Recommendations:

For low Mach and low Reynolds number flows, the incompressible setup available in SU2 may be more appropriate. It is therefore recommended to perform a similar study using the SU2 incompressible formulation. For further details, refer to:

https://su2code.github.io/docs_v7/Theory/#incompressible-navier-stokes

References:

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